



Technology advancement is driving electric vehicle adoption

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Edited by Marilyn Brown, Georgia Institute of Technology, Atlanta, GA; received November 13, 2022; accepted April 11, 2023

Electric vehicle sales have been growing rapidly in the United States and around the world. This study explores the drivers of demand for electric vehicles, examining whether this trend is primarily a result of technology improvements or changes in consumer preferences for the technology over time. We conduct a discrete choice experiment of new vehicle consumers in the United States, weighted to be representative of the population. Results suggest that improved technology has been the stronger force. Estimates of consumer willingness to pay for vehicle attributes show that when consumers compare a gasoline vehicle to its battery electric vehicle (BEV) counterpart, the improved operating cost, acceleration, and fast-charging capabilities of today's BEVs mostly or entirely compensate for their perceived disadvantages, particularly for longer-range BEVs. Moreover, forecasted improvements of BEV range and price suggest that consumer valuation of many BEVs is expected to equal or exceed their gasoline counterparts by 2030. A suggestive market-wide simulation extrapolation indicates that if every gasoline vehicle had a BEV option in 2030, the majority of new car and near-majority of new sport-utility vehicle choice shares could be electric in that year due to projected technology improvements alone.

electric vehicles | consumer choice | preferences | vehicle technology | discrete choice experiment

Technology development and consumer adoption of battery electric vehicles (BEVs) are among the greatest contributors to uncertainty in the energy efficiency and carbon intensity of future passenger transportation (1). While BEVs have historically been a small percentage of the vehicle market, the pace of recent technological change in BEVs has been rapid, with battery costs alone dropping by a factor of ten from 2010 to 2021 (2). The average range of BEVs has increased by 200%, while efficiency has increased by 15% (3), and the number of BEV offerings has grown dramatically (4). At the same time, consumer exposure to BEVs socially has likely increased as the number of these vehicles on the road has grown (3, figure 4.14), major policies have pushed electric vehicles to the forefront of political debates (5, 6), and major automakers have pledged to solely provide electric vehicles in the near future (7). Coffman et al. (8, p.86) provide a review of literature showing that social interactions can influence BEV adoption. The questions of how the consumer probability of choosing a BEV has changed over time, what is driving changes in consumer choices, and how much BEV market share may increase in the future have important implications for automotive technologies and policies. For example, California recently passed emission standards that effectively ban the sale of new cars that run on gasoline only (5), and the U.S. Environmental Protection Agency recently released proposed emission standards estimated to require 2/3 of new vehicle sales to be electric by 2032 (9). The viability of these emissions targets and policies depends on whether technological improvements, changes in consumer preferences, or both, can generate large increases in BEV market share in the near future.

This paper examines consumer choices of plug-in electric vehicles*, including BEVs and plug-in hybrid electric vehicles (PHEVs) relative to conventional gasoline vehicles (CVs). We focus on how consumer demand for plug-in electric vehicles has been changing over time, accounting for how the technology has improved and allowing for changing preferences. We field a discrete choice experiment where US new vehicle consumers choose among potential vehicle options, mimicking the process of comparing vehicles on an automaker's website. The results of the experiment are compared to a companion discrete choice experiment that was conducted in 2012 to 2013 in order to examine changes in consumer vehicle choices over time (10). In both cases, the weighted respondent pool is a representative sample of new vehicle buyers in the United States.

*The term "plug-in electric vehicle" (PEV) refers to vehicles that acquire some or all of their propulsion energy from an external electricity source (usually the power grid), including 1) battery electric vehicles (BEVs) and 2) plug-in hybrid electric vehicles (PHEVs), which use a blend of electricity and petroleum for propulsion including extended-range electric vehicles (EREVs). PEVs do not include traditional hybrid electric vehicles (HEVs) (such as the Toyota Prius) that do not plug in.

Significance

This study provides insight into current and past drivers of battery electric vehicle (BEV) adoption as well as the future for BEV adoption in the United States—one of the most influential and uncertain aspects of transportation greenhouse gas emissions. Using large-scale, nationally representative survey data, we show that technology improvements have been a stronger force increasing the market share of BEVs in recent years than changes in consumer preferences. Further, we present evidence indicating that BEVs could constitute the majority or near-majority of cars and SUVs by 2030, given widespread BEV availability and technology trends. This work is relevant to policymakers, vehicle manufacturers, and researchers as they consider the trends influencing decarbonization of light-duty transportation.

Author contributions: C.R.F., K.T.G., J.J.M., and K.S.W. designed research; performed research; contributed new reagents/analytic tools; analyzed data; and wrote the paper.

Competing interest statement: K.T.G. has served as a consultant for the California Air Resource Board, Center for Applied Environmental Law and Policy (environmental NGO) and the Toyota Research Institute. This research was supported in part by a grant from the Center for Applied Environmental Law and Policy.

This article is a PNAS Direct Submission.

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This article contains supporting information online at [http://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2219396120/-DCSupplemental](https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2219396120/-DCSupplemental).

Published May 30, 2023.

The results show that advances in BEV technology—in particular increases in range and reductions in the BEV price-premium—have driven substantial increases in consumer choices of BEV cars and SUVs over their conventional gasoline vehicle counterparts. Estimates of consumer willingness to pay for vehicle attributes show that any perceived disadvantages of BEVs relative to gasoline vehicles are often compensated by the BEV's improved operating cost, acceleration, and fast-charging capabilities, particularly for BEVs with a longer range. This, combined with technology cost reductions that are expected to reduce the BEV price premium, implies that forecasts of technology improvements are especially important for projecting consumer demand for plug-in electric vehicles going forward. In contrast, while it is possible that consumer preferences may have changed, we do not find statistically significant changes in these preferences over the past decade.

Using the resulting estimates in consumer choice simulations, we find that for today's passenger cars that offer both gasoline and BEV powertrain options, the average premium consumers are willing to pay for a BEV over a gasoline version of the same vehicle ranges from $-\$12,000$ to $\$6,600$, depending on the vehicle. We report all monetary values in year 2022 USD using the consumer price index, unless otherwise noted. Accounting for expected improvements in BEV range and price by 2030, this WTP shift to a range of $-\$5,300$ to $\$8,000$. For SUVs, we see a similar trend, with willingness to pay a premium for BEVs shifting from $-\$8,900$ to $-\$8,400$ today to a range of $-\$6,000$ to $-\$4,400$ by 2030.

Simulating a future scenario where every conventional gasoline vehicle has an available BEV counterpart and BEV technology improvements follow projections from the National Academies (1) with supply adequate to match demand at projected prices, we estimate that BEVs would make up the majority of new car sales and near-majority of new SUV sales by 2030.

Given the limited historical evidence available to understand mainstream consumer preferences for plug-in electric vehicles, we draw upon carefully constructed discrete choice survey experiments with randomized vehicle profiles, using a choice-based conjoint design. Specifically, the results are derived from two discrete-choice survey experiments run eight years apart and designed to be as comparable as possible, while accounting for changes in the automobile market. The surveys are of a representative sample of new vehicle buyers in the United States.[†] The contribution of this paper is both the second survey, conducted in 2020 to 2021, and a comparison of this second survey to the previous survey performed in 2012 to 2013 (10).[‡]

This work contributes to a broad literature on the consumer preferences for plug-in electric vehicles. Classic work on vehicle preferences focuses on a static equilibrium setting where the market and technology is not changing (e.g., see the studies reviewed in refs. 11 and 12). Some work attempts to incorporate a time dimension in estimating consumer preferences by asking individuals to consider their own future decisions (13, 14), but there is little empirical work on the question of how consumer preferences and demand for emerging technologies like plug-in

electric vehicles might be changing over time due to technology improvements or changes in preferences.

We identify only three related studies in the peer-reviewed literature that attempt to explore trends in consumer preferences for plug-in electric vehicles. In the first notable contribution, Carley et al. (15) examined the changes in stated intention to purchase a plug-in electric vehicle between 2011 to 2017 from potential new vehicle purchasers in the largest 21 US cities, finding that American consumers were more intent on purchasing plug-in vehicles in 2017 relative to 2011. Second, Jenn et al. (16) examined changes throughout 2010 and 2017 in California plug-in electric vehicle purchasers' ratings of the importance of various incentives, such as rebates, on their decision to purchase a plug-in electric vehicle, finding that adoption of these vehicles has become more dependent on incentives over time. Finally, Kurani (17) uses survey data up to 2017 to show that the distribution of individuals considering electric vehicles has not dramatically shifted. These studies lay the groundwork for understanding consumer preferences relating to plug-in electric vehicles but answer different questions in a notably different automobile market than today due to the rapidly changing technology. Furthermore, they do not explore the degree to which consumer willingness to trade off relevant vehicle attributes associated with electrification (e.g., range, operating cost, price, etc.) may have changed over time due to technology improvements or other factors and what this could imply for the sales of new vehicles in upcoming years. Our study sheds light on these questions.

Scope of Study

Our data collection approach was designed to examine changes in preferences of mass-market US consumers—meaning consumers representative of US new vehicle purchasers. Because early-adopters' preferences differ from mass-market preferences (18–23), revealed-preference (RP) data, such as historical sales, may not provide good estimates of mass-market consumer preferences, so we instead leverage stated-preference (SP) data. There are many pros and cons of using SP data over RP data (24, 25, pp. 21–24; Table 1). In particular, the use of SP data allows us to present respondents with electric vehicles that are not yet available on the market (24, pp. 22–23). For instance, a BEV passenger car with 300 miles of range could be presented with a purchase price of $\$17,000$. Such a vehicle is not currently available in the market, although it is anticipated to be available within the next 5 to 10 y (1). Further, we are able to account for how large changes in fuel and energy costs (beyond the variation observed in recent fuel prices) impact preferences over time. This capability of SP data allows us to analyze how future changes in technologies, as well as fuel and energy prices, may influence consumer vehicle purchases. The use of SP data also allows us to conduct a controlled experiment that would be prohibitive to conduct in the marketplace. Use of controlled experiments enables the modeler to 1) observe all of the same attributes observed by the respondent (unlike RP data, where consumers make purchases while observing attributes that are not available to the modeler); 2) observe the full choice set (unlike RP data, where the modeler does not typically know what alternatives were available at the time of purchase or which the consumer considered or was even aware of); and 3) avoid confounding (unlike RP data, where strong correlations, such as between EV technology and efficiency, can make it difficult to identify whether consumers are buying EVs because they are electric or because they are efficient). Finally, SP data avoid supply-side issues that interfere with demand-side estimates,

[†]The sample was collected to represent mainstream new vehicle consumers. Consistent with this population, many of the consumers in our sample do not have experience with electric vehicles.

[‡]Because the second study was performed during the COVID-19 pandemic, we asked respondents how the pandemic impacted them and reestimate results excluding respondents who were impacted in ways that may have influenced their preferences for vehicles. We do not find any statistically significant differences in results. Details are provided in *SI Appendix*.

such as reduced vehicle availability during the COVID-19 pandemic.

The key criticism of SP data is that it may not reflect the decisions consumers would make in the marketplace when they must commit large amounts of money, as is the case with purchasing a new vehicle. Aiming to mitigate this concern, we incorporate multiple features into the survey design that tend to improve the ability for survey responses to reveal comparable preferences as when making true purchase decisions (26). First, we use a discrete choice survey design that mimics the experience of consumers comparing vehicle specifications and prices during the purchase decision and simultaneously limits the cognitive burden on respondents to improve the reliability of responses. Second, we use the range of vehicle performance specifications and prices of vehicles available in the market so that if any anchoring effects exist in the survey (27), they replicate anchoring effects that would be present in the market during the purchasing decision. Third, we explain to respondents that their responses will be used to inform automaker decisions on vehicle offerings because it has been shown that SP survey respondents that believe that their responses will have an impact on decision-makers tend to give responses that are consistent with choices that have financial consequences (26).

Our survey was conducted from December 2020 to September 2021. After some introductory questions and information, survey respondents chose 1) either passenger cars or SUVs, 2) vehicle size, and 3) a given aesthetic among a set of vehicle image options.[§] The image selected was held fixed for the remainder of the survey to represent the chosen vehicle to avoid the potential for respondents to conflate vehicle attributes with presumed styling or vehicle class differences. Respondents were then shown a series of fifteen choice tasks. In each choice task, respondents were asked to select their preferred option among three vehicle profiles that varied in price, powertrain type (i.e., conventional gasoline vehicle, gasoline-electric hybrid, PHEV, and BEV), operating cost, 0 to 60 mph acceleration time, range, whether the vehicle has fast-charging capability (if it is a BEV), and the brand country-of-origin (e.g., American, German, Japanese, Korean, Chinese). Values for these attributes were varied randomly across vehicle profiles and choice tasks to systematically and causally test the effect of varying vehicle attributes on consumer choice. The survey design is almost identical to that of ref. 10 to enable comparison, with updates made to reflect the range of attributes available in the 2020 to 2021 vehicle market. Additional details on the survey design are presented in *SI Appendix*.

In our sample, there are 734 car-buyer and 862 SUV-buyer survey responses from people who had intentions of purchasing a car within the next two years or had purchased a car within the prior year of when the survey was fielded, requirements also used in ref. 10 so as to ensure comparability. Respondents were recruited using both Amazon's Mechanical Turk (mTurk) and Dynata. mTurk was chosen to replicate data collection from the study conducted in 2012 to 2013 in order to investigate changes in consumer preferences over time. Dynata was chosen because it includes older and higher-income respondents, which are underrepresented by the mTurk sample and improves coverage for generating a representative sample. We weight the respondents in our analysis to ensure representativeness with the US new car and new SUV buying

population. Alternative data weighting results are available in *SI Appendix*.

Results

How Are BEVs Valued Relative to Conventional Vehicles? We begin the presentation of our results with a set of head-to-head comparisons, focusing on vehicle models that offer both a conventional and electric powertrain option.[¶] These comparisons provide a clean way to illustrate relative consumer preferences without conflating unobserved attributes (styling, interior design, etc.). We look at these head-to-head comparisons using WTP values estimated from the 2021 survey data (full results are available in *SI Appendix*). We also evaluate expected technology progression from ref. 1 for a hypothetical near-future vehicle. These comparisons set the stage for forward-looking simulations across the entire fleet and help identify the main drivers of our results.

Fig. 1 shows a head-to-head comparison in a waterfall chart over time. The figure compares the Nissan Leaf to a close conventional counterpart built on the same platform, the Nissan Versa. The y-axis is the WTP for the BEV relative to the conventional vehicle. Fig. 1*A* shows the results for the 2013 Leaf and the 2013 Versa. Fig. 1*B* shows the 2022 Leaf and the 2022 Versa for a present-day comparison. Fig. 1*C* shows a hypothetical Leaf with 300 miles of range and the 2022 Versa for a comparison of what might occur in the near future, based on projections from ref. 1. Lines are included in the first two panels to indicate the actual BEV price premium, both with and without a \$7,500 federal tax credit.

In Fig. 1*A*, we observe that in 2013, relative to the Versa, the Leaf had a shorter (75 mile) range, which reduced the WTP, a lower average operating cost (*SI Appendix* for calculations), which increased WTP, and a slower acceleration, which lowered WTP. On net, the difference in the WTP for the Leaf versus the Versa, on average, is $-\$14,000$, and the price premium was almost $+\$20,000$ before the tax credit. This gap is consistent with the relatively low choice share of the 2013 Leaf compared to the Versa.

Fig. 1*B* shows a distinctly different picture for model year 2022, given the same consumer preferences. The range of the Leaf is 149 miles, which produces a less negative WTP relative to the Versa. The operating cost, acceleration, and BEV fast-charging capability all increased the WTP. On net, the 2022 Leaf WTP is nearly on par with the Versa. Thus, if the two were priced the same, we should expect to see similar choice share of both. The 2022 Leaf has a $\$12,000$ price premium over the Versa before the tax credit, and we observe lower choice share of the Leaf than the Versa, as expected. But comparing panels (*A*) and (*B*) shows a substantial change in the overall net WTP due to technology changes alone (cases including changes in preferences are available in *SI Appendix*).

Fig. 1*C* provides insight into what might happen in upcoming years with improved battery technology that allows for longer-range vehicles at a lower cost. For this scenario, we hold the operating cost, acceleration, and fast-charging capability fixed but assume a 300-mile range and no price premium for the BEV, based on projections from (1, figure 5.37). With these changes, the net WTP for the BEV is above the zero price

[§]Note, we consider only the car- and SUV-buyer markets, so these results cannot be extrapolated to the entire US vehicle market. However, these two vehicle classes make up a large majority of the overall American vehicle market (3, figure 3.2), making these results informative with respect to changes in the overall vehicle market.

[¶]Nearly all vehicle specifications were collected from *Edmunds*. Acceleration measures were collected from *Carindigo* for the most recent model year available relative to the model year listed. Press releases (found [here](#) and [here](#)) from Nissan give the MSRP for the 2013 Nissan Leaf and Versa. The base trim level is used in all comparisons unless otherwise noted.

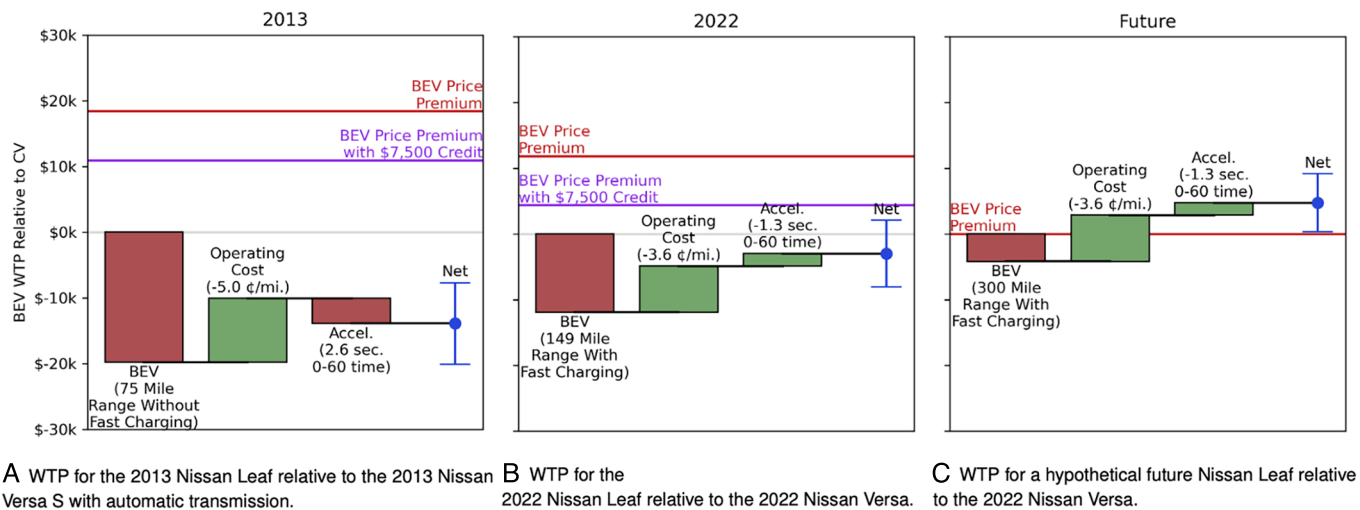


Fig. 1. Head-to-head charts showing WTP for attributes for the Nissan Leaf BEV relative to those of the Nissan Versa gasoline vehicle (model year 2013 in A, 2022 in B, and hypothetical future Nissan Leaf in C), which is built on the same platform, using consumer preference data from the 2021 survey. Horizontal lines show the price premiums associated with the electric vehicle with and without the federal BEV tax credit applied. Error bars denote ± 2 standard errors.

premium, suggesting higher share of choices for the Leaf than the Versa.

We repeat the head-to-head comparison in Fig. 1 for all plug-in electric vehicles available today that offer a gasoline powertrain option, and Fig. 2 summarizes the results. The first row of panels are cars and the second row shows SUVs. To

simplify, we present the net WTP numbers after accounting for all attributes (range, operating cost, etc.). These net WTP estimates are presented in blue. In red, we present the BEV price premium. We include estimations using our survey data across all vehicle model years, including extrapolations out to 2030 that hold preferences constant at 2021 levels, extend the range of all

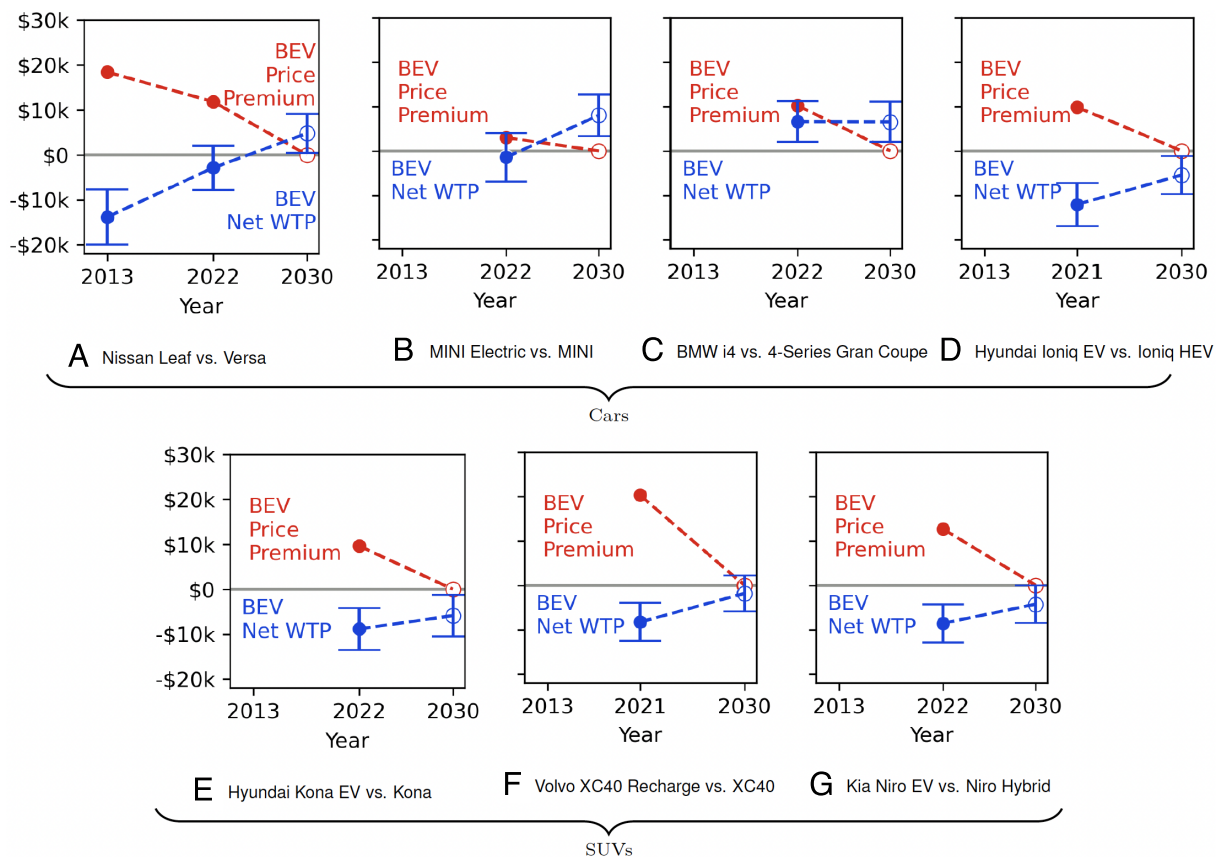


Fig. 2. Car and SUV head-to-head comparisons over time (Panels A–G refer to a different head-to-head comparison). Red points denote the price premium of the BEV relative to the comparable gas-powered vehicle. Blue points denote the net willingness to pay (WTP) of the BEV relative to the comparable gas-powered vehicle. Car and SUV net WTP calculated using the 2021 study mixed logit model for car-buyers and SUV-buyers respectively. Error bars denote ± 2 standard errors. Every panel shares the same axes.

BEVs to 300 miles, and assume no price premium (acceleration assumptions were informed by figure 4 in ref. 28, and range and price assumptions are informed by the projections figure 5.37 in ref. 1).

The pattern in the results in Fig. 2 is consistent: for all of the 2021 comparisons, the net WTP is well below the price premium, but by 2030 the expected improved range and reduced price premium produce WTP estimates comparable to or greater than the price premium. Specifically, for the Leaf, Mini, and BMW i4, the forecasted improvements to 2030 have each BEV preferred to its gasoline counterpart, while for the Kona and XC40, the two powertrains are comparable. For the Ioniq and Niro (which are compared to an HEV counterpart), the projected WTP is still below the price premium. As before, a key finding is that changes in attributes are a clear force leading to greater WTP for BEVs, even with consumer preference parameters held constant.

Willingness to Pay Estimates. The WTP estimates represent the average value (price-equivalency) that consumers place on changes in vehicle attributes. Consistent with prior work, we find that if all vehicle attributes (range, operating cost, acceleration, brand, etc.) are identical across powertrain types, consumers prefer conventional gasoline vehicles over BEVs and PHEVs, on average. However, consumers significantly value improvements in attributes that plug-in electric vehicles offer, such as reduced operating cost, that have been shown to counteract the preference for gasoline vehicles. Our results estimate that car (SUV) buyers are willing to pay, on average, \$4,160 (\$8,700) more for a gasoline vehicle than a BEV of the same range with fast-charging capability when the vehicles have identical other attributes, and they are willing to pay an additional \$1,470 (\$1,440) per second of reduction in 0 to 60 time; \$1,960 (\$1,490) per 1¢/mi reduction in operating cost; \$5,120 (\$7,010) per 100 miles of additional range; and \$4,140 (\$4,110) for BEV fast-charging capability.

We compare WTP estimates across the new and old datasets using Wald tests for equality of means and equality of the full set of distribution parameters in *SI Appendix*. We find no robust evidence to suggest that mean consumer preferences for BEVs or BEV range have shifted over time for either car or SUV buyers (additional specifications are discussed in *SI Appendix*). It should be clarified that, although we cannot identify changes over time, this does not mean there have not been any. Rather, there is not sufficient evidence to reject our null hypothesis that preferences have not changed. Additionally, this finding does not imply that consumer preferences will necessarily remain static in the future, as consumers gain exposure to electric vehicles.

It should be noted that there is an inherent limitation with the 2015 study dataset in that it has fewer observations than the 2021 study dataset. It is therefore possible that additional changes in consumer preferences have occurred during this period but were not large enough to be identified with statistical significance given the size of the 2015 sample.

Consumer Choice Implications. The head-to-head comparisons are valuable to illuminating how changes in BEV technology and cost can drive increased BEV WTP and choice shares in a concrete set of comparisons that reasonably hold factors outside the scope of our analysis constant. However, ultimately it is the overall market shares that matter. We construct future scenarios and run market-wide simulations to highlight what our estimation results might imply for the entire vehicle market if

each conventional vehicle were to offer a BEV powertrain option. This assumption may be reasonable for 2030, given that there are dozens of planned BEV offerings by automakers (29). We base our simulations on data from the 2022 CAFE Compliance Model (30). It is important to emphasize that vehicle sales, and resulting market shares, result from the interaction of supply and demand, and our study assesses only demand. Specifically, our market simulations assume sufficient supply of electric vehicles at prices estimated by a recent National Academies report (1).

We focus on two scenarios. The first is a hypothetical current vehicle market where every conventional vehicle has a BEV counterpart, which is useful to show how more BEV offerings could change BEV share with today's technology. For each CV, we assume a BEV counterpart that is on par with offerings that exist today: it has a 200-mile range and a 48% price premium. The second scenario is a hypothetical future market (model year 2030) where each conventional vehicle has a BEV counterpart with 300 miles of range and a 0% price premium, representing technology projections from a recent National Academies report (1). Finally, we incorporate any unobserved attributes by calibrating alternative-specific constants (ASCs) to match mean market share with the market share data (30), and we assume BEV counterparts will have identical unobserved attributes and, thus, identical ASCs. Details for the simulation can be found in *SI Appendix*.

Fig. 3 presents the results of this simulation. We observe that BEV offerings result in higher simulated car and SUV BEV market shares relative to today's market shares (which result from interactions of supply and demand and for which there are far fewer BEV options). Furthermore, our results suggest that the simulated BEV market share for both cars and SUVs would increase to roughly half the market by 2030, assuming widely available and comparable BEV offerings. These findings highlight that expected technological improvements are key to the adoption of BEVs.

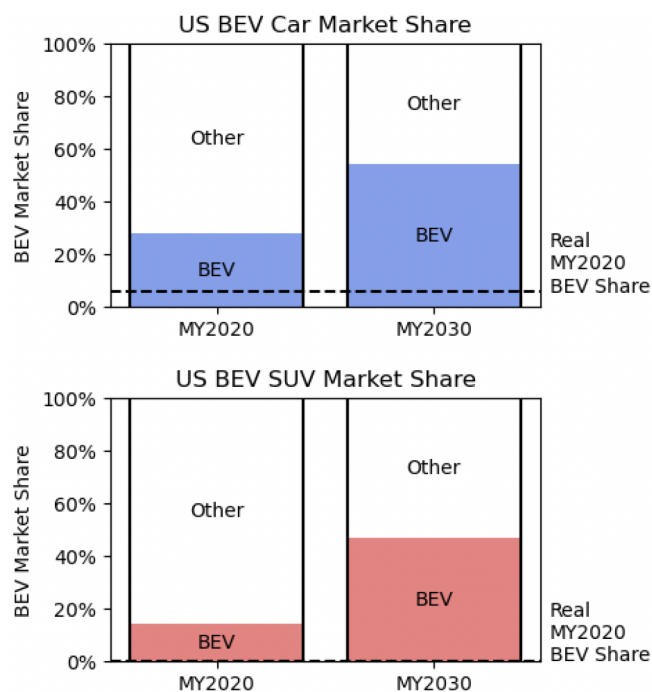


Fig. 3. Hypothetical market-wide simulation for model year 2020 and 2030 where all internal combustion engine and hybrid engine vehicle models have a battery electric vehicle option associated with them.

Sensitivity Analysis. We perform a set of sensitivity analyses to characterize the robustness of our results. One key question stems from the weighting of our survey data to accurately represent the vehicle-buying population. The 2015 companion study (10) uses MTurk data supplemented with data from the Pittsburgh Auto Show (an event designed to attract primarily ordinary car buyers, rather than enthusiasts), while our current survey uses data from MTurk and Dynata. To assure that these slightly different data sources do not affect our results, we reweight and reestimate both models using only data from Mechanical Turk respondents (*SI Appendix*). We find that the only robust, identifiable differences among car buyers in these subsamples are the mean and distributional difference tests for operating costs and the American brand. For SUV buyers, there are no robust differences identified.

In a second sensitivity analysis, we examine the implications of varying WTP for operating cost to reflect estimates from the literature in place of our estimates (31–34). Specifically, we apply lower and upper bounds from the literature of fixed WTP for a 1¢/mile increase in operating cost of –\$232 and –\$1,378 for car buyers and –\$250 and –\$1,438 for SUV buyers. We find that our market-wide simulation results remain fairly robust, with substantial BEV adoption at either end of the range of operating cost WTP (*SI Appendix* for more details) and relatively modest changes in magnitude. Our market simulations are also similarly robust to alternative assumptions regarding fuel prices and acceleration improvements for BEVs.

Discussion and Implications for Technology and Policy

Our findings have important implications for vehicle policies as well as BEV technology development. Understanding the trajectory of consumer willingness to adopt BEVs is crucial for the effectiveness of many recent and proposed policies that aim to encourage vehicle electrification (see the wide range of policies reviewed throughout (35) as well as (5, 36)). Our results imply that the likelihood of consumers purchasing BEVs has grown over time because of technological improvements that have increased range and (in many cases) reduced price premiums of particular BEV models relative to their gasoline counterparts. This trend improves the viability and reduces the costs of regulations that encourage electrification, such as stronger greenhouse gas emission standards and zero-emission vehicle regulations.

Should technological projections hold, our simulation results suggest that, when vehicles are offered with both gasoline and BEVs powertrain options, the BEV may capture about half of powertrain choices by 2030 even without BEV purchase incentives. This is in line with forecasts by financial and consulting companies that rely on expert predictions (37, 38), and they suggest a need to prepare BEV-related infrastructure (e.g., charging stations, sufficient power generation and transmission, etc.) for a pending increase in adoption. These results also lend additional evidence to the optimism about transitioning the automotive industry to focus more on BEV production in the coming decade, and they suggest that technology progress projections are key for future BEV adoption projections used in policy planning and cost–benefit analyses. Importantly, we do not model supply-side factors that could affect market outcomes, and our results assume substantial BEV model availability.

Additionally, our results provide a direction for BEV technology development that can increase consumer adoption. Our consumer choice estimates underscore the potential importance of increasing BEV range. Most vehicles with a range of at least 300 miles were valued by consumers equivalently or more than

their conventional gasoline vehicle counterparts. While BEVs have some disadvantages, such as longer recharging times than it takes to fill up a gasoline vehicle, our results show that these disadvantages are made up for, on average, by lower operating costs and fast-charging capability as long as range is sufficiently long. The results also suggest that further improving the efficiency of BEV powertrains to deliver faster acceleration and/or lower operating costs can help increase consumer adoption.

Interestingly, we find little evidence of major changes in underlying consumer preferences independent of vehicle technology. One might speculate that growing awareness of plug-in electric vehicles between 2013 and 2021 would have increased the likelihood of consumers purchasing these vehicles even if the technology remained constant over time. While our results do not rule out that some change in consumer preferences occurred, we do not find statistically significant changes in consumer valuation of plug-in electric vehicles over time once the specific technology and vehicle attributes are controlled for. This could be interpreted to imply that consumer awareness efforts are less effective than technology development or, conversely, that inadequate resources have been devoted to consumer awareness efforts in the past decade. It is possible that consumer preferences could change in the future as larger numbers of consumers gain experiences with PEVs (39).

There are some limitations to our study worth mentioning. We use a stated preference approach because historical data on plug-in electric vehicle sales is scant and largely limited to early adopters, who may differ considerably from mass-market consumers (40). We take several precautions in our discrete choice experiment to minimize potential stated preference biases, but respondents may nevertheless make different choices in hypothetical choice scenarios than in purchase scenarios. Our sensitivity analysis suggests that our general findings are robust to variation in model parameters based on the literature. As more plug-in electric vehicles are adopted over the coming years, revealed preference work could complement our findings.

Our consumer choice results also assume that the availability of BEVs is ubiquitous and consumers can just as easily find and purchase these vehicles through automotive retailers and dealerships as conventional gasoline vehicles. This is not the case in some parts of the US today, but it may be true in 2030. We also focus our study on passenger cars and SUVs for greater comparability to previous work, but further work on pickup trucks is warranted, especially given that pickup trucks are now 14% of new vehicle sales in the United States. (3, p. 16). We conclude by noting that there is room for much more work exploring preferences for plug-in electric vehicles, especially across geography, buyer characteristics (income, race, etc.), and in the used vehicle market.

Materials and Methods

In our analysis, we use a random coefficients modeling framework that allows for flexible substitution patterns between vehicle offerings and thus better characterizes consumer preferences than simpler approaches. Specifically, our approach models consumer choice with a random-coefficient (mixed) logit utility over the attributes of the vehicles (41). We estimate the model in willingness-to-pay (WTP) space (42), allowing us to interpret the coefficients as WTP parameters directly, while relaxing the limitations of the common independence assumption for the distribution of the coefficients (43).

Vehicle Choice Model. We model consumer i 's utility for vehicle alternative j as follows:

$$u_{ij} = \lambda \left(\alpha_i a_j + \omega_i c_j + \beta_i b_j + \rho_i r_j + \eta_i f_j + \delta_i^T \mathbf{x}_j - \rho_j \right) + \epsilon_{ij} \quad [1]$$

where $\alpha_j \in \mathbb{R}$ is the WTP per unit increase in vehicle acceleration time $a_j \in \mathbb{R}$ (0-60 mph time in seconds), $\omega_j \in \mathbb{R}$ is WTP per unit increase in vehicle operating costs $c_j \in \mathbb{R}$ (cents/mile), $\rho_j \in \mathbb{R}$ is WTP per unit increase in BEV range $r_j \in \mathbb{R}$ (miles), $\beta_j \in \mathbb{R}$ is WTP for a BEV powertrain $b_j \in \{0, 1\}$ relative to the baseline gasoline vehicle (with identical range), $\eta_j \in \mathbb{R}$ is WTP for BEV fast-charging capability $f_j \in \{0, 1\}$, and $\delta_j \in \mathbb{R}^n$ is a vector of WTP coefficients for the vector of remaining indicators $\mathbf{x}_j \in \{0, 1\}^n$ for gasoline-electric hybrid and PHEV powertrains as well as vehicle brand variables and fast-charging indicator variables. $\lambda \in \mathbb{R}$ is a scaling factor that identifies the magnitude of the price signal relative to the normalized standard deviation of the error term. p_j is the price of the vehicle, and ϵ_j is a type I extreme value error term.

The estimated parameters are λ and the respondent population mean and standard deviation for each of the WTP random coefficients $\alpha, \omega, \beta, \rho, \eta$, and δ . For tractability, we adopt the common assumption that all WTP parameters are independent and normally distributed across the population. Use of the WTP space avoids the conflation of taste heterogeneity and scaling effects otherwise implied by this assumption (43).

We use robust standard errors and account for multiple choice observations from each respondent. The model thus assumes that the error terms are independent only across individuals. In addition to the preferred random-coefficients (mixed) logit model above, we estimate alternative model specifications, which are described in *SI Appendix*. Survey design and replication

instructions, including selection of attributes and levels, are also detailed in *SI Appendix*. Finally, we would like to note that the entire study was approved by the Carnegie Mellon University Institutional Review Board, and that all survey respondents gave informed consent to participate in this study.

Data, Materials, and Software Availability. Anonymized survey data can be accessed at the following DOI: <https://doi.org/10.5281/zenodo.7901414> (44). Code for the analysis can be found on Github at: <https://github.com/cfrorsythe/DrivingElectrificationReplication> (45).

ACKNOWLEDGMENTS. The authors would like to acknowledge funding for this project from the Center for Applied Environmental Law and Policy. They would also like to acknowledge helpful comments from Elaine Buckberg, Kristin Brainerd, the editor, and three anonymous reviewers.

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